

Conjecture 00000005120: same spectrum, exponentially different condition numbers of the matrix inverse, separated by two Jordan structures

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2026-10-05

Statement

English. Definition: The spectrum and the function condition number are two layers of sensitivity. Conjecture: There exist two matrices with the same spectrum but exponentially different matrix-function condition numbers, and the separation is realized by an explicit pair of Jordan structures. (The Chinese text says the same: two matrices with the same spectrum whose matrix-function condition numbers differ exponentially, the separation realized by an explicit pair of Jordan structures.)

Verdict: proved. The statement is existential. “Exponentially different” needs a growing parameter, so we give an explicit family indexed by the dimension $n \geq 2$. Everything below is the single Lean theorem `Conjecture5120.conjecture_5120`, checked against Mathlib.

1 Definitions and reading

- *Matrices and spectrum.* Real $n \times n$ matrices. The spectrum is Mathlib’s `spectrum R M` (the μ with $\mu I - M$ not invertible). We also prove equality of characteristic polynomials, so the eigenvalues agree with algebraic multiplicity (over $\mathbb{C}$ as well, since the polynomial splits).
- *Jordan structures.* $J_n(\lambda) = \lambda I + N$, where N has ones on the superdiagonal (`jordanBlock`), and the scalar Jordan matrix λI with n blocks of size 1 (`scalarJordan`).
- *Matrix function.* $f(x) = 1/x$, whose matrix function is $f(X) = X^{-1}$ (`Mathlib.Ring.inverse`).
- *Matrix norm.* $\|M\|_\infty = \max_i \sum_j |M_{ij}|$, the operator norm induced by the sup norm on $\mathbb{R}^n$ (`Mathlib.Matrix.linfTyOpNormedRing`; Lean type `Mat n`).
- *Condition numbers* (the standard definitions, as in Higham, *Functions of Matrices*, Ch. 3). The absolute condition number of f at X is the operator norm of the Fréchet derivative, $\text{cond}(f, X) = \|L_f(X)\| = \max_{Z \neq 0} \|L_f(X)Z\|/\|Z\|$ (Lean `cond`, defined as `fderiv`’s operator norm). The relative condition number is $\text{cond}_{\text{rel}}(f, X) = \text{cond}(f, X) \|X\|/\|f(X)\|$ (Lean `condRel`). Wikipedia (“Condition number”) states the same: in several variables the condition number uses the norm of the derivative, “ $J(x)$ denotes the Jacobian matrix of partial derivatives of f at x , and $\|J(x)\|$ is the induced norm on the matrix.”

The Fréchet derivative of inversion is Mathlib’s `fderiv_inverse`: $L_{\text{inv}}(X)Z = -X^{-1}ZX^{-1}$.

2 Theorem

For $n \geq 1$ let $A_n = \frac{1}{2}I_n$ (n Jordan blocks of size 1) and $B_n = J_n(\frac{1}{2}) = \frac{1}{2}I_n + N_n$ (one Jordan block of size n).

Theorem 1 (`conjecture_5120`). *For every $n \geq 2$:*

1. $\chi_{A_n} = \chi_{B_n} = (X - \frac{1}{2})^n$ and $\sigma(A_n) = \sigma(B_n) = \{\frac{1}{2}\}$;
2. A_n and B_n are not similar (different Jordan structures);
3. $\text{cond}(\text{inv}, A_n) = 4$ and $\text{cond}(\text{inv}, B_n) \geq 4^n$, so $\text{cond}(\text{inv}, B_n) \geq 4^{n-1} \text{cond}(\text{inv}, A_n)$;
4. $\text{cond}_{\text{rel}}(\text{inv}, A_n) = 1$ and $\text{cond}_{\text{rel}}(\text{inv}, B_n) \geq 2^n/4$.

3 Proof

(1) Same spectrum. Both matrices are upper triangular with every diagonal entry $\frac{1}{2}$ (`A_upper`, `B_upper`), so `Mathlib's charpoly_of_isUpperTriangular` gives $\chi = (X - \frac{1}{2})^n$ for both (`charpoly_A`, `charpoly_B`). For a field, $\mu \in \sigma(M)$ iff μ is a root of χ_M (`mem_spectrum_iff_isRoot_charpoly`); the only root of $(X - \frac{1}{2})^n$, $n \neq 0$, is $\frac{1}{2}$ (`spectrum_of_charpoly`).

(2) Different Jordan structures. For $n \geq 2$, $N_n \neq 0$ (its $(0, 1)$ entry is 1). If $B_n = P A_n P^{-1}$ then, since A_n is scalar, $P A_n P^{-1} = A_n$, so $N_n = B_n - A_n = 0$, a contradiction (`not_similar`).

(3) Absolute condition numbers. $A_n^{-1} = 2I$, so $L(A_n)Z = -4Z$ and $\|L(A_n)\| = 4$: the bound $\|4Z\| \leq 4\|Z\|$ gives ≤ 4 , and any $Z \neq 0$ gives ≥ 4 (`cond_A`). For B_n the inverse is explicit:

$$(B_n^{-1})_{ij} = c(i, j) = \begin{cases} (-1)^{j-i} 2^{j-i+1}, & i \leq j, \\ 0, & i > j, \end{cases}$$

proved by checking $B_n C = I$ entrywise (`key`, `B_mul_Binv`): row i of $B_n C$ is $\frac{1}{2}c(i, j) + c(i+1, j)$, which is 1 for $j = i$ and, for $j = i + e + 1$, $\frac{1}{2}(-1)^{e+1}2^{e+2} + (-1)^e 2^{e+1} = 0$. Take $Z = E_{n-1,0}$ (a single 1 in position $(n-1, 0)$), with $\|Z\|_{\infty} \leq 1$ (`norm_single_le`). Then $(B_n^{-1} Z B_n^{-1})_{0, n-1} = c(0, n-1)^2 = (2^n)^2 = 4^n$ (`entry_calc`). Since every entry is bounded by the ∞ -norm (`entry_le_norm`), $4^n \leq \|L(B_n)Z\| \leq \|L(B_n)\| \|Z\| \leq \|L(B_n)\|$ (`cond_B`).

(4) Relative condition numbers. $\|\frac{1}{2}I\| = \frac{1}{2}$ and $\|2I\| = 2$ (`norm_scalar`), so $\text{cond}_{\text{rel}}(\text{inv}, A_n) = 4 \cdot \frac{1}{2}/2 = 1$ (`condRel_A`). For B_n : $\|B_n\| \geq |(B_n)_{00}| = \frac{1}{2}$ (`norm_B_ge`), and $|c(i, j)| \leq 2^{j+1}$ gives every row sum of B_n^{-1} at most $\sum_{j < n} 2^{j+1} = 2^{n+1} - 2$, so $0 < \|B_n^{-1}\| \leq 2^{n+1}$ (`norm_Binv_le`, `norm_Binv_pos`). Hence $\text{cond}_{\text{rel}}(\text{inv}, B_n) \geq 4^n \cdot \frac{1}{2}/2^{n+1} = 2^n/4$ (`condRel_B`). $\square$

4 Scope and conventions

- The matrix function is fixed as $f(x) = 1/x$; the conjecture asks for some matrix-function condition number, and inversion is the classical one. The exponential gap is not claimed for every f . For $f = \exp$, for example, the same pair is *not* separated exponentially (remark, not formalized): $\|e^{sN_n}\|_{\infty} \leq e^s$ for $0 \leq s \leq 1$, and the integral formula $L_{\exp}(X)Z = \int_0^1 e^{sX} Z e^{(1-s)X} ds$ then keeps $\text{cond}(\exp, B_n) \leq e^{3/2}$ for every n .
- “Exponentially different” is read as a ratio that grows exponentially in a parameter of an explicit family. Here the parameter is the dimension n , and the ratios are at least 4^{n-1} (absolute) and $2^n/4$ (relative). The eigenvalue $\frac{1}{2}$ is fixed.

- The matrix norm is the ∞ -operator norm. Other norms change the constants. The lower bound $\text{cond}(\text{inv}, B_n) \geq 4^n$ uses only one entry of $B_n^{-1} Z B_n^{-1}$ and $\|E_{n-1,0}\|_\infty = 1$. The Lean statement is proved for the ∞ -norm only.
- The spectrum is taken over $\mathbb{R}$; equal characteristic polynomials give equal complex spectra and multiplicities as well.

5 Lean formalization and axiom audit

File `lean/Conjecture5120/Basic.lean` (Lean 4, Mathlib v4.33.1, only `import Mathlib`; 403 lines). Main theorem:

```
theorem conjecture_5120 (n : Nat) (hn : 2 <= n) :
  A n = scalarJordan n (1/2) /\ B n = jordanBlock n (1/2) /\
  (A n).charpoly = (B n).charpoly /\
  spectrum R (A n) = {1/2} /\ spectrum R (B n) = {1/2} /\
  (not exists P : (Matrix (Fin n) (Fin n) R)^x, B n = P * A n * P^-1) /\
  cond n (inv n) (A n) = 4 /\ 4^n <= cond n (inv n) (B n) /\
  4^(n-1) * cond n (inv n) (A n) <= cond n (inv n) (B n) /\
  condRel n (inv n) (A n) = 1 /\ 2^n / 4 <= condRel n (inv n) (B n)
```

Here `cond n f X := ||fderiv R f X||` (operator norm of the Fréchet derivative on `Mat n`, the real $n \times n$ matrices with the ∞ -operator norm), `condRel n f X := cond n f X * ||X|| / ||f X||`, and `inv n := Ring.inverse`. `lake build` succeeds, and `#print axioms Conjecture5120.conjecture_5120` reports `[propxt, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom.